

Bulk Ising susceptibility: a candidate natural-boundary proof

Expert brief for manuscript v0.1-rc4; Lean formalization / post-audit release v0.2.1

Dehao Lin, School of Physics, Sun Yat-sen University

ORCID: 0009-0001-4551-8490

lindh9@mail2.sysu.edu.cn

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Repository: <https://github.com/LeoLam233/ising-bulk-natural-boundary>

The object and the gap

The accompanying manuscript presents a candidate complete proof that $|s| = 1$ is a natural boundary for the zero-field, isotropic, infinite square-lattice **bulk susceptibility**, continued from the low-temperature + boundary-condition state. This candidate is prepared for public mathematical review. For $s = \sinh(2\beta_{\text{phys}}J)$, the normalized exterior germ inherited from real $s > 1$ is

$$\mathcal{X}(s) = \beta_{\text{phys}}^{-1} \chi(s) = 1 - \mathcal{M}^2(s) + 2\mathcal{M}^2(s) \sum_{\substack{N \geq 2 \\ N \text{ even}}} T_N(s), \quad \mathcal{M}^2(s) = (1 - s^{-4})^{1/4}.$$

The response sums connected correlations over the full lattice and retains the infinite particle expansion.

Nickel (1999, 2000) identified fixed-form-factor singularities whose union is dense on the unit circle. Orrick et al. (2001) gave amplitude-based evidence against cancellation in the full sum, explicitly short of a proof. Tracy–Widom (2014) proved smooth extension away from Nickel points at each fixed even order. Their diagonal-susceptibility (2013) and Toeplitz-sum (2015/2017) works provide rigorous noncancellation precedents. The gap addressed here is control of the *entire higher-particle tail in the bulk response* near the boundary, so that it cannot cancel the first singular contribution.

Two estimates that carry the conclusion

For each fixed point s_* in the selected cyclotomic family of §3, let $N_0 = 2p$ be its first even order, with $p \geq 11$ prime, $k = N_0^2/2 - 1$, and $s_\varepsilon = (1 + \varepsilon)s_*$. The two claimed estimates, as $\varepsilon \downarrow 0$, are

$$\begin{aligned} \partial_s^k T_{N_0}(s_\varepsilon) &= 2L_\beta \varepsilon^{-1/2} + o(\varepsilon^{-1/2}), \quad L_\beta \neq 0, \\ \sum_{\substack{N > N_0 \\ N \text{ even}}} |\partial_s^j T_N(s_\varepsilon)| &= o(\varepsilon^{-1/2}), \quad 0 \leq j \leq k. \end{aligned}$$

The first supplies a nonzero singular carrier; the second makes the entire higher-particle tail too small to cancel it. The first coefficient belongs to the whole form factor, not just a local model. The tail covers every higher even order and the full angular domain. Constants may depend on the fixed point and finite k . With the stated lower-order bounds, normal convergence and symmetries, these estimates yield the claimed full natural boundary (§9).

The proposed bulk implementation

The technical claims are a dense uniquely-first point family, a whole-form-factor first-amplitude calculation, an exact weighted contour/Stokes decomposition, protected parameter disks, and differentiated full-tail/coarea estimates. The noncancellation principle is established prior work. Contour partitions, Vandermonde analysis, grouping and Hadamard/factorial budgets also have Tracy–Widom precedents (§1.1). No exhaustive priority clearance is claimed.

External inputs. Palmer–Tracy’s (1981) low-temperature Fredholm correlation representation, as used in the published Tracy–Widom (2014), Appendix 1; Yang’s (1952) magnetization; and Tracy–Widom’s fixed-even-order non-Nickel smoothness. None supplies the bulk infinite-tail theorem. Sources and conventions are in manuscript §1.1, §2 and Appendix F.

Proof architecture

The two branches of the argument meet in the noncancellation step:

$$\begin{aligned} \text{Dense uniquely-first family} &\longrightarrow \text{nonzero whole first singularity,} \\ T_N = F_N + K_N + S_N &\longrightarrow \text{uniform differentiated full tail,} \\ \text{Both estimates} &\longrightarrow \text{dense boundary noncancellation} \\ &\longrightarrow \text{natural boundary.} \end{aligned}$$

Here F_N is the final selected-contour integral, K_N the retained original contribution, and S_N the weighted Stokes correction. The tail argument combines protected-disk estimates, pair/Pfaffian suppression and differentiation/coarea bounds near collisions.

Suggested independent reading

Each interface below can be read separately as a first check.

- 1. First carrier and amplitude — §4, Appendix B.** Does the fixed y -only localization, followed by full x residues and the mean-angle deformation, give the coefficient of the *whole* form factor? Check that the two chart coefficients add without cancellation and that the complement and lower derivatives remain controlled.
- 2. Weighted contours and pair control — §§5–8.** Check the exact identity $T_N = F_N + K_N + S_N$ on the actual coupled support, including its Stokes correction. Do the protected parameter disks, regularized complete pairs and same-group contraction survive branch and endpoint degeneracies? Does the selected-contour Pfaffian budget retain every differentiation and integration cost?
- 3. High-order differentiation and coarea — Appendices D–E.** Does the Lie-derivative recurrence include derivatives of previously generated coefficients and real cutoffs? Check puncture flux at each stage, the real coarea Jacobian and multiplicity, and the order of constants and quantifiers needed for the uniform estimates.
- 4. Complete tail closure — §9, Appendix G.** Do the thirteen sectors W1–W13 cover all higher even orders and all $0 \leq j \leq k$, including transitions between supports and particle-number windows? Verify that their bounds sum to the stated smaller-order tail.

Specific feedback sought

A concrete failing configuration, an unjustified implication or uniformity step, or a missed close precedent would be especially useful. **A limited reading of one interface is valuable; a full refereeing commitment is not presumed.**

Current verification status. The internal form-factor and higher-tail proof chain is formalized in Lean 4. An independent adversarial audit of frozen v0.2.0 gave `PASS_WITH_NONBLOCKING_ISSUES`, finding no blocking or load-bearing defect. Release v0.2.1 applied targeted post-audit remediation without changing the core natural-boundary theorems. The exact v0.2.1 commit passed the final targeted regressions and controls, a clean full build, a project-wide axiom audit of 9,200 safe declarations, 14 endpoint checks, fresh kernel replay, and full and smoke CI. E1/E2/E3 remain explicit external literature inputs: the low-temperature correlation representation, Yang’s spontaneous-magnetization formula, and Tracy–Widom’s fixed-even-order non-Nickel smoothness. This remains an AI-assisted candidate proof: no independent human expert validation or peer review is claimed.

Section and appendix pointers refer to the accompanying manuscript v0.1-rc4; full bibliographic details are in its reference list.

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